



Central Assessment of Longitudinal Knee X-rays for Quantitative JSW

1. Overview	1
1.1 SAS dataset	1
1.2 Contents of dataset	1
1.3 Condition	2
1.4 Variables and reading methods	2
2. Methods specific to Project 16	6
2.1 Image type	6
2.2 Time points	6
2.3 Measurement methods	6
2.4 Variables	6
2.5 Sample	7
3. References	8

1. Overview

1.1 SAS dataset

Name: **kXR_QJSW_DuryeaXX** (XX identifies the time point)

Display label: **kXR quantJSW (Duryea)**

1.2 Contents of dataset

This dataset contains central longitudinal measurements of joint space width (JSW) and related parameters from serial OAI knee x-rays, performed in the laboratory of Dr. Jeff Duryea at Brigham and Women's Hospital in Boston, MA. <http://www.spl.harvard.edu/pages/People/duryea>. The data from more than one project by this vendor may be contained in the dataset. A "reading project" consists of a particular sample of knees read over a specific period of time using a single protocol.

NOTE: The previous release of these datasets contained JSW measurements for knees of all OAI participants with longitudinal knee x-rays at any time point up to the 48-month visit who had radiographic knee OA, along with a large number of knees without radiographic knee OA and the Non-Exposed Control (NEC) subcohort, as well as measurements of any 72-month and 96-month visit knee x-rays for those same subjects (as long as their knee x-rays were not end-stage (JSN=3) at their last visit from baseline thru 48-months). In this dataset (versions 0.8, 1.8, 3.7, 5.5, 6.5, 8.2, and 10.2) an additional group of randomly selected knees without radiographic knee OA at any time point have had their JSW measurements performed and added to these datasets.

Each dataset contains the reading data for a single visit, or time point. The corresponding longitudinal data from reading projects by this vendor for other time points are in separate datasets. For example, the data from this vendor's reading of baseline x-rays are in a dataset ending in "00" (i.e., kXR_QJSW_Duryea00), while the corresponding data from this vendor for the 12-month visit x-ray readings are in a dataset with the same name but ending in "01" (i.e., kXR_QJSW_Duryea01), and so on. (See the "[VisitPrefixDefinitions.pdf](#)" document for a guide to visit numbering.) To compare values of a variable across time points from a given project by this vendor, or to calculate change scores, users will need to merge the datasets for the various time points.

IMPORTANT: All users are strongly encouraged to read the *Overview and Description of Central Image Assessments* document carefully before attempting to use this dataset!



For comprehensive and detailed information about the structure and contents of the central image assessment datasets, general guidelines for defining change between time points and duration of follow-up, and strategies for merging datasets for analysis, please see the “*Overview and Description of Central Image Assessments*” document.

1.3 Condition

- Known data errors: problems/cautions for use are listed by variable in the “Release Comments”, which can be found in the various *kXR_QJSW_DuryeaXX_Comments.pdf* files.
- Dataset strengths/weaknesses:
 - o Data are expected for all participants included in a project sample. If expected data do not exist for a knee, SAS special missing values are assigned to denote why the data were not obtained.
 - o The dataset only contains a single row of data (1 record) for a given knee, but other OAI datasets may contain multiple records per knee and this needs to be taken into account when merging with other datasets. Please see the “*Overview and Description of Central Image Assessments*” document for more information on merging.
 - o To examine longitudinal changes in JSW for a specific project, records corresponding to that project in the *kXR_QJSW_Duryea00* datasets are used for baseline visit values, and then longitudinal changes from those values are found by using values of the same variable in the follow-up *kXR_QJSW_DuryeaXX* datasets which have matching values for ID, SIDE and READPRJ (although for the current *kXR_QJSW_DuryeaXX* datasets, READPRJ only takes a single value, but this might change in future releases of data).
 - o Project 16 provides lateral compartment JSW measurements. Measurement of JSW in this compartment has not been studied to the same extent as in the medial tibio-femoral compartment, and the results from this project for lateral compartment JSW should be considered experimental until their usefulness has been examined.
- Overlap between reading projects in this dataset:
 - o Unlike previous versions of these datasets, only one record per knee exists.
 - o Data from previous projects of JSW x-ray readings (Projects 02, 05, 13) have been unreleased and replaced by Project 16 data.
 - o Separate datasets (*kXR_QJSW_Rel_DuryeaXX*) of reliability data have been released and replace data from Project 13.

1.4 Variables and reading methods

See the dataset documentation file *kXR_QJSW_DuryeaXX_Contents.pdf* for a complete list of all the variables in the dataset, their SAS variable names, descriptive variable labels and attributes.

Key variables measured for all projects by this vendor include:

- Minimum JSW in the medial tibiofemoral compartment



- JSW at fixed locations (medial to lateral) in the tibiofemoral joint

The image analysis, measurement techniques and variables are similar in the different projects by this vendor, and are outlined below. Any modifications for a specific project and project specific variables are described in the section related to that project, including:

- Measurements made at each time point
- Criteria for selecting the knees to analyze

Quantitative measurements of tibiofemoral JSW are made at specific anatomic locations within a knee joint. These publications give more details about the methods and measurement performance:

- Neumann G, Hunter D, Nevitt M, Chibnik LB, Kwoh K, Chen H, Harris T, Satterfield S, Duryea J, For the Health ABC Study. Location specific radiographic joint space width for osteoarthritis progression. *Osteoarthritis Cartilage*. 2009 Jun; 17(6): 761-765. PMID: 19073368 <http://dx.doi.org/10.1016/j.joca.2008.11.001>
- Duryea J, Li J, Peterfy CG, Gordon C, Genant HK. Trainable rule-based algorithm for the measurement of joint space width in digital radiographic images of the knee. *Med Phys*. 2000 Mar;27(3):580-91. PMID: 10757609 <http://dx.doi.org/10.1118/1.598897>

In general, a customized software tool developed by Dr. Duryea is used to provide a measurement of JSW on serial fixed-flexion knee radiographs from the OAI. Figure 1 shows the output of the software. The software automatically delineates the margin of the femoral condyle and the tibial plateau. The software automatically checks to ensure that the joint is fully delineated in the measurement region. The reader also verifies that the joint margins are fully delineated and there is consistency between the different visits. If these checks fail the reader can re-analyze the image using tools that make a semi-automated correction of the software generated margins. In extreme cases the reader can manually redraw the contours.

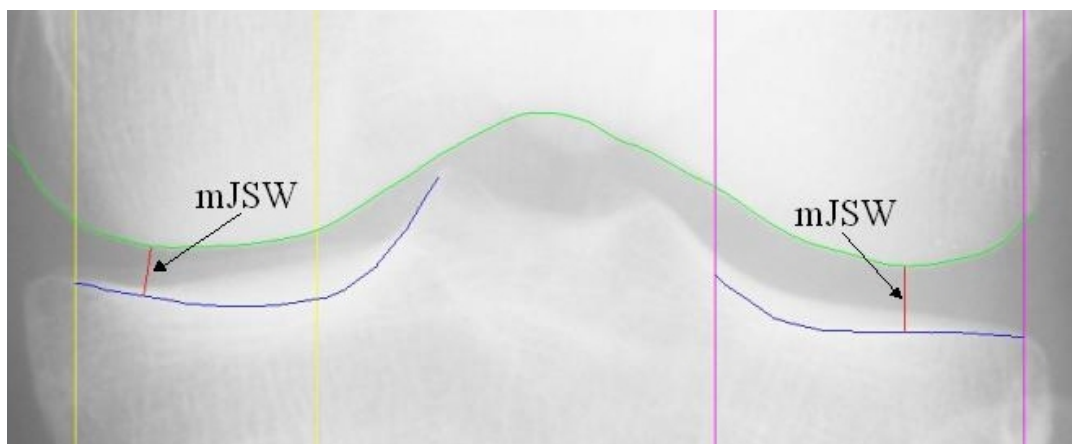


Figure 1. Output of the software on a digital knee radiograph. mJSW is the minimum JSW in a compartment. We provide mJSW for the medial compartment only.

Measures of the medial compartment minimum JSW (mJSW) and fixed location JSW (JSW(x)) are made from digitized knee images at each time point. The mJSW is defined at the location of the smallest distance between the femoral and tibial joint margins. The software also measures the JSW at fixed locations (JSW(x)) along the joint. An anatomical coordinate system is defined, which permits an objective



determination of the x-location. (Figure 2) The x-axis is placed automatically by the software and the y-axis and a line defined as $x = 1$ are placed manually by a reader. A study of longitudinal knee radiographs suggested that $x = 0.2$ to $x = 0.275$ may be the optimal range for measuring medial JSW(x). (1) (Figure 3)

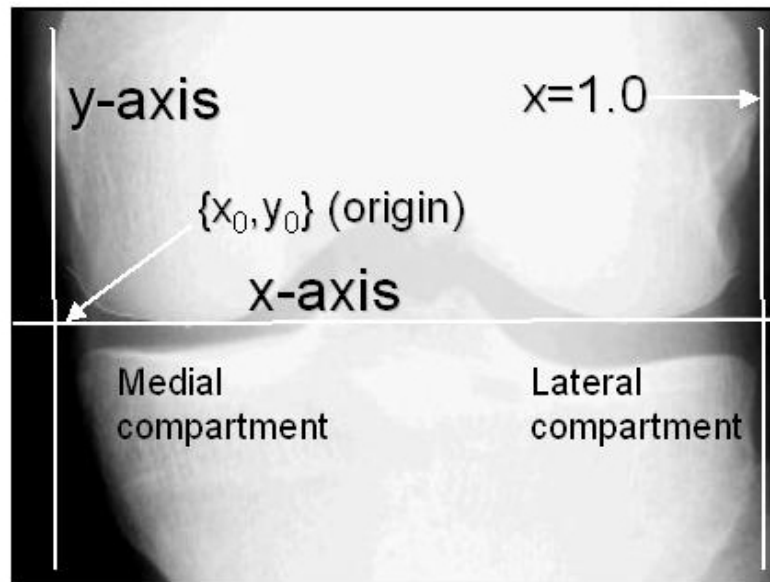


Figure 2. Landmarks and definition of coordinate system.

JSW measurements taken at $x > 0.5$ are for the lateral tibio-femoral compartment. We provide measurements taken from $x=0.7$ to $x=0.9$ at increments of 0.025, although further work is needed to determine the optimal region over which lateral compartment JSW measurements reflect underlying structural alterations in that side of the joint.

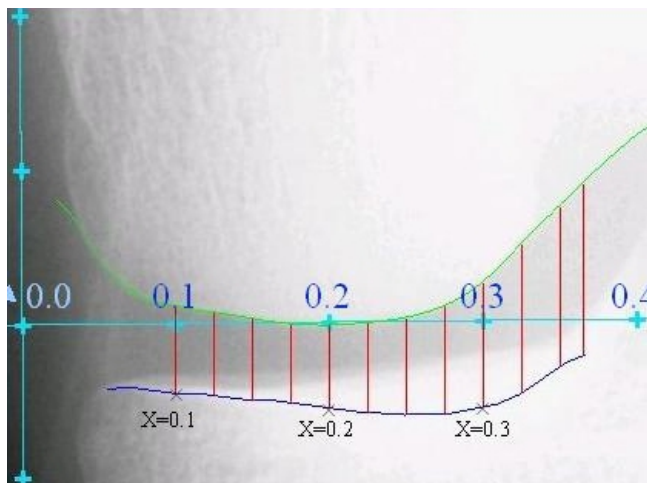


Figure 3. Measurement of JSW(x). Measurements from $x=0.150$ to $x=0.300$ are provided in increments of 0.025.

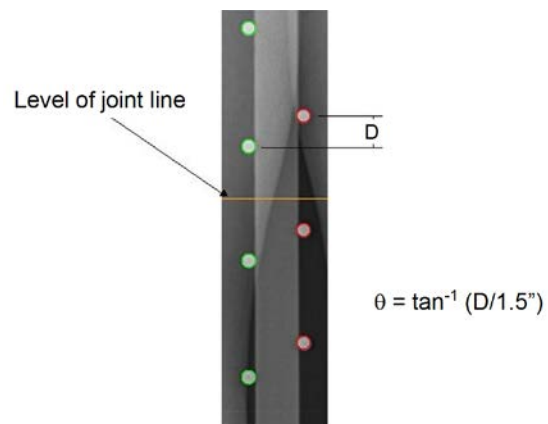


Figure 4. Formula for calculating the approximate beam angle (θ) at the level of the joint line from the vertical spacing (D – in inches) between 2 matching metal beads on the Synaflexer frame.



The protocol for acquiring the fixed flexion radiographs requires that the x-ray beam makes an angle of 10 degrees downwards at the level of the joint line. To check that this is the case, the difference in relative heights of two matched beads from the Synaflexer™ Frame can be used to estimate the beam angle for an image. Figure 4 shows an example of the metal beads and the formula used to estimate the angle.

Missing data can occur for a variety of reasons. In the individual kXR_QJSW_DuryeaXX datasets, SAS special missing values are used to indicate the following:

- .P if data is missing due to a prosthesis/knee replacement
- .T if data is missing due to technical reasons
- .A if the data is not expected

Missing data can also occur after merging data from different visits. For example, for Project 16, a participant may have V00JSW..., V03JSW..., V05JSW... and V06JSW...scores but if that person did not attend their 12-month visit, they would have no data in the kXR_QJSW_Duryea01 dataset, but in a merged dataset containing longitudinal data from baseline, 12-month and 24-month visits, that participant would have missing values for their V01JSW...variables.

JSW measurements in the dataset are expressed to 3 decimal places to help minimize the accumulation of rounding errors during certain calculations, such as additional magnification correction using the Synaflexer™ Frame. However, we recommend reporting all JSW results to 2 decimal places in publications.

Tibial Plateau Alignment

Measurements of alignment of anterior and posterior rims of the tibial plateau are not provided, but there is a variable VxxTPCFDS available for each knee with JSW measurements at each time point, which can be used to assess consistency of positioning between visits. Although use of this variable is currently experimental, we currently believe that changes in this measurement of more than +/-2 mm between visits may indicate that knee position has varied significantly between the visits, and a value of this variable larger than 6.5 mm may indicate that a single visit has a poorly positioned knee. In future releases of OAI data we may have updated information about use of this variable.



2. Methods specific to Project 16

2.1 Image type:

Fixed-flexion bilateral knee radiograph. See the “*Radiographic (X-ray) Manual*” document for the acquisition protocol.

2.2 Time points:

Baseline, 12-month, 24-month, 36-month, 48-month, 72-month, and 96-month visits.

2.3 Measurement methods:

The manual and semi-automated aspects of the reading were done viewing the images paired but blinded to chronological order, with paired images viewed simultaneously. Any newly measured knees had baseline thru 48-month visits measured paired by blinded to chronological order, and once those measurements were complete, 72-month and 96-month visit images were measured, blinded to chronological order, but where the reader had the ability to compare anatomical landmark positions between the newly measured visits and the previously measured visits.

2.4 Variables:

For this project:

- Minimum JSW in the medial tibio-femoral compartment.
- The position (x co-ordinate) of the location of the minimum medial JSW.
- Medial compartment fixed location JSW(x) measured at 7 locations between $x = 0.150$ and $x = 0.300$, in increments of 0.025 units (see Figure 3 above).
- Lateral compartment fixed location JSW(x) measured at 9 locations between $x = 0.700$ and $x = 0.900$ in increments of 0.025 units.
- A measurement (“tibial rim distance”), taken at the center of the medial compartment, of the distance between (a) the anterior or posterior margin of the tibial plateau (whichever is closest to the femoral condyle), and (b) the tibial margin of the joint space outlined during the JSW measurement process. This measurement is NOT an assessment of the alignment the anterior and posterior rims of the tibial plateau, but changes in this measurement can be used to assess consistency of radiographic positioning between time points.
- The width of the femoral condyles, used to define the anatomical locations (see Figure 2 above). This defines the width of the femoral condyles from similar anatomical locations across each visit for a knee. It could in theory be used to normalize JSW measurements for variations in magnification.
- An estimate of the beam angle of the x-rays to the film/detector at the level of the joint line based on the positions of the 2 metal beads closest to the level of the joint space (see Figure 4 above for the method of calculation).
- The pixel size (in mm) used to convert from pixel units into distance in millimeters.
- Various flag variables for each knee measured to indicate criteria which may be used to determine the reliability of any cross-sectional JSW measurements, or longitudinal changes in JSW measurements:
 - o Minimum medial compartment JSW set to zero because reader judges bone-on-bone (as opposed to actually performing a measurement).
 - o Minimum medial compartment JSW is measured, but is not a local minimum.
 - o Medial and/or lateral compartment JSW cannot be measured due to poor positioning/image quality.
 - o Incomplete delineation of medial and/or lateral compartment – some fixed location JSW measurements are missing.
 - o Inconsistent positioning for at least one time point because the “tibial rim distance” varies by more than 2mm over all the time points measured. Examine the actual changes in the measurement to determine which visits pairs have inconsistent positioning.



- o Margin of lateral compartment JSW is not the same as the rim of the lateral tibial plateau closest to the femoral condyles. This situation can occur due to radiographic positioning, or simply due to the anatomy of a participant's knee, and is indicated because it is unclear how this situation affects lateral compartment JSW.

2.5 Sample:

The following table gives some demographic information about the participants with data currently released for Project 16:

Project 16 sample: Distribution of Race by Sex

	White or Caucasian	Non-White	Total
Male	1,198	228	1,426
Female	1,543	499	2,042
Total	2,741	727	3,468*

*Race data missing for 1 participant

Project 16 sample: Distribution of Age by Sex

	Age (years)				Total
	45 to 49	50 to 59	60 to 69	70 to 79	
Male	163	513	375	376	1,427
Female	199	663	711	469	2,042
Total	362	1,176	1,086	845	3,469



3. References

1. Neumann G, Hunter D, Nevitt M, Chibnik LB, Kwoh K, Chen H, Harris T, Satterfield S, Duryea J, For the Health ABC Study. Location specific radiographic joint space width for osteoarthritis progression. Osteoarthritis Cartilage. 2009 Jun; 17(6): 761-765. PMID: 19073368
<http://dx.doi.org/10.1016/j.joca.2008.11.001>
2. Duryea J, Li J, Peterfy CG, Gordon C, Genant HK. Trainable rule-based algorithm for the measurement of joint space width in digital radiographic images of the knee. Med Phys. 2000 Mar;27(3):580-91. PMID: 10757609 <http://dx.doi.org/10.1118/1.598897>